Unit 1 Progress Check: MCQ Part A

1. A student wants to create an algorithm to calculate the average (mean) of two quiz scores. Which of the following algorithms can be used to determine the correct average?

(A)

Step one:	Add the two quiz scores.
Step two:	Multiply the result of step one by 2 to determine the average.

(B) ✓

Step one:	Add the two quiz scores.
Step two:	Divide the result of step one by 2 to determine the average.

(C)

Step one:	Subtract the lower quiz score from the higher quiz score.
Step two:	Multiply the result of step one by 2 to determine the average.

(D)

Step one:	Subtract the lower quiz score from the higher quiz score.
Step two:	Divide the result of step one by 2 to determine the average.

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Answer B

Correct. An average is the sum of the list of numbers divided by how many numbers are in the list. In this case, there are two numbers, so the sum is divided by 2.

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2. A student wants to create an algorithm to calculate the combined total area of two rectangles. Which of the following algorithms can be used to perform this calculation?

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(A)

Step 1:	Multiply the length of the first rectangle by the length of the second rectangle.
Step 2:	Multiply the width of the first rectangle by the width of the second rectangle.
Step 3:	Add the result of step 1 to the result of step 2.

(B) ✓

Step 1:	Multiply the length of the first rectangle by the width of the first rectangle.
Step 2:	Multiply the length of the second rectangle by the width of the second rectangle.
Step 3:	Add the result of step 1 to the result of step 2.

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(C)

Step 1:	Add the length of the first rectangle to the width of the first rectangle.
Step 2:	Add the length of the second rectangle to the width of the second rectangle.
Step 3:	Multiply the result of step 1 by the result of step 2.

(D)

Step 1:	Add the length of the first rectangle to the length of the second rectangle.
Step 2:	Add the width of the first rectangle to the width of the second rectangle.
Step 3:	Multiply the result of step 1 by the result of step 2.

Answer B

Correct. Step 1 calculates the area of the first rectangle by multiplying its length by its width. Step 2 calculates the area of the second rectangle by multiplying its length by its width. Step 3 calculates the combined area by adding the areas determined in steps 1 and 2.

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3. A student wrote a program to calculate the average of several numeric scores. The program compiles without error but produces incorrect results when it runs. The student suspects there is a logic error in the code.

Which of the following strategies is most likely to help identify the logic error?

(A) Changing random parts of the code and rerunning the program until the error is resolved

(B) Inserting print statements throughout the code to check the calculations at various points in the execution ✓

(C) Relying on the compiler's error messages to detect the logic error

(D) Running the program multiple times without making changes to see if the error corrects itself

Answer B

Correct. Inserting print statements in code can show what the code is doing as it executes. This can reveal the point when an error or incorrect result is introduced.

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4. Consider the following variable declaration.

```
int x;
```

Which of the following values can be stored in the variable x?

(A)

0.5

(B) ✓

1

(C)

true

(D)

false

Answer B

Correct. An int variable can be assigned an integer value such as 1.

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5. Consider the following variable declaration.

```
boolean b;
```

Which of the following values can be stored in the variable b?

(A)

-1

(B)

0

(C)

1.0

(D) ✓

false

Answer D

Correct. A boolean variable can be assigned only the values true or false.

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6. A code segment will use variables to represent a student's age, in years, and whether the student has a driver's license. Which of the following variable declarations are most appropriate for the code segment?

(A)

```
boolean age;

double hasLicense;
```

(B)

```
boolean age;

boolean hasLicense;
```

(C) ✓

```
int age;

boolean hasLicense;
```

(D)

```
int age;

double hasLicense;
```

Answer C

Correct. Ages are typically represented using non-decimal numbers, so an `int` variable is most appropriate. For whether or not something is true, like whether a student has a driver's license, a `boolean` variable is most appropriate.

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7. Consider the following code segment.

```
int a = 1;
int b = 2;
int c = 3;
int d = 4;
double x = a + b * c % d;
```

What is the value of x after executing the code segment?

(A)

1.0

(B)

2.5

(C) ✓

3.0

(D)

7.0

Answer C

Correct. The multiplication b * c is performed first and evaluates to 6. The remainder operation 6 % d is performed next and evaluates to 2. The addition operation a + 2 is performed last and evaluates to 3.

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8. Which of the following code segments will print the word Hello?

(A) ✓

```
System.out.print("Hello");
```

(B)

```
System.out.print(Hello);
```

(C)

```
"System.out.print(Hello)";
```

(D)

```
System.out.print(He);

System.out.print(llo);
```

Answer A

Correct. The code segment correctly prints the string literal "Hello".

9. Consider the following statement.

```
System.out.print("AP \n Computer \n Science \n A \n rocks");
```

Which of the following best describes the behavior of this statement?

(A) It prints five lines of text, each separated by a line break indicated by the escape sequence \n. ✓

(B) It prints four lines of text because the escape sequence \n appears four times in the string literal.

(C) It prints one line of text because the printed value is a single string literal.

(D) It prints nothing due to a syntax error.

Answer A

Correct. Each newline escape sequence (\n) inside the string literal causes a line to print.

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10. Consider the following code segment.

```
boolean a = true;
boolean b = false;
boolean temp = a;
a = b;
b = temp;
System.out.println(a);
System.out.println(b);
```

What is printed as a result of executing this code segment?

(A)

true

true

(B)

true

false

(C) ✓

false

true

(D)

false

false

Answer C

Correct. This code segment swaps the values of a and b. It initially assigns true to a and false to b. Next, the value of a, which is true, is assigned to temp. Next, the value of b, which is false, is assigned to a. Next, the value of temp, which is true, is assigned to b. The final values of a and b are false and true.

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11. The following code segment contains an error.

```
double pi = 3.14;    // line 1
pi = 3.14159;        // line 2
double other = null; // line 3
double example;      // line 4
example = pi;        // line 5
```

Which of the following best describes the error in this code segment?

(A) In line 2, pi is assigned a value after it was already assigned a value in line 1.

(B) In line 3, null cannot be assigned to a non-reference type. ✓

(C) In line 4, example should be assigned a value.

(D) In line 5, a variable cannot be assigned to another variable.

Answer B

Correct. The keyword null can only be assigned to a reference type, not to a primitive type like double.

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12. The volume of a cylinder is equal to the height times the area of the circular base. The area of the circular base is equal to π (pi) times the square of the radius.

The following code segment is intended to compute and print the volume of a cylinder with radius r and height h. Assume that the double variables r, h, and pi have been properly declared and initialized.

```
/* missing code */
System.out.print(volume);
```

Which of the following **cannot** be used to replace /* missing code */ so that the code segment works as intended?

(A) ✓

```
double baseArea;

double volume;

pi * r * r = baseArea;

baseArea * h = volume;
```

(B)

```
double baseArea = pi * r * r;

double volume = baseArea * h;
```

(C)

```
double volume = pi * r * r ;

volume = volume * h;
```

(D)

```
double volume = pi * r * r * h;
```

Answer A

Correct. The code segment uses incorrect assignment statements. The expressions pi * r * r and baseArea * h should be on the right side of the = rather than the left side.